

Verifier Certificate 001

“I know that I know that I am an AMLD”

Status. Conditional verifier-facing proof skeleton. It is not yet executable until the predicates and certificate rules named below are frozen in the verifier language.

Settlement target.

$$\tau_A(t) := K_{A,t+1}(K_{A,t} \text{ AMLD}(\langle A \rangle)).$$

Requested terminal settlement: P .

Trusted checker: V .

Scope of the claim. VC001 is a conditional theorem schema. In addition to the certified structural inputs below, it assumes the epoch/accessibility convention in L07a and the two epistemic-domain premises in L07b. In particular, the second occurrence of K is not obtained automatically from AMLD status alone.

Definitions

L01. Define the AMLD federation by

$$A := (M_P, M_R, M_I, M_C, M_V; B, S, I).$$

Justification: definition of the AMLD federation.

L02. Define its canonical formal name by

$$\langle A \rangle := \text{Code}(M_P, M_R, M_I, M_C, M_V; B, S, I).$$

Justification: canonical naming definition.

L03. Define the finite structural formula

$$\Phi_A := \left(\bigwedge_{j \in \{P, R, I, C, V\}} \text{ATP}(\langle M_j \rangle) \right) \wedge \text{Comp}(A).$$

Here $\text{Comp}(A)$ abbreviates the finite certified structural requirements involving the P/R/I/C/V roles, BANK, scheduler/control, interfaces, verifier relation, and the other frozen AMLD composition requirements.

L04. Define AMLD membership structurally by

$$\text{AMLD}(\langle A \rangle) \iff \Phi_A.$$

Justification: structural definition of AMLD.

L05. Introduce the abbreviation

$$\sigma_A := \text{AMLD}(\langle A \rangle).$$

L06. For a certified epistemic domain E , define verifier-grounded AMLD knowledge by

$$K_{A,t}\varphi \iff [\varphi \in E \wedge \exists r (r \in \text{View}_A(t) \wedge \text{Witness}_V(r, \varphi))].$$

Justification: verifier-grounded accessible-warrant definition of AMLD knowledge.

L07. Define the operational, time-explicit target by

$$\tau_A(t) := K_{A,t+1}(K_{A,t}\sigma_A).$$

L07a. Epoch/accessibility convention. $\text{View}_A(t)$ is the verifier-accepted record view at the closure of epoch t . A record deposited and accepted during epoch t before closure is therefore in $\text{View}_A(t)$. The next accepted meta-record, deposited during epoch $t + 1$ before its closure, is in $\text{View}_A(t + 1)$.

L07b. Epistemic-domain premises. For this certificate, assume

$$\sigma_A \in E \quad \text{and} \quad K_{A,t}\sigma_A \in E.$$

The second premise is exactly the one level of receipt-reflection closure required for the target $\tau_A(t)$; no unrestricted positive introspection principle is assumed.

Certified structural inputs

L08. Provide a verifier-acceptable proof certificate for the proof-role ATP:

$$\pi_P : \text{ATP}(\langle M_P \rangle), \quad V(\text{ATP}(\langle M_P \rangle), \pi_P) = 1.$$

L09. Provide a verifier-acceptable proof certificate for the refutation-role ATP:

$$\pi_R : \text{ATP}(\langle M_R \rangle), \quad V(\text{ATP}(\langle M_R \rangle), \pi_R) = 1.$$

L10. Provide a verifier-acceptable proof certificate for the independence-role ATP:

$$\pi_I : \text{ATP}(\langle M_I \rangle), \quad V(\text{ATP}(\langle M_I \rangle), \pi_I) = 1.$$

L11. Provide a verifier-acceptable proof certificate for the contradiction/integrity-role ATP:

$$\pi_C : \text{ATP}(\langle M_C \rangle), \quad V(\text{ATP}(\langle M_C \rangle), \pi_C) = 1.$$

L12. Provide a verifier-acceptable proof certificate for the verifier-role ATP:

$$\pi_V : \text{ATP}(\langle M_V \rangle), \quad V(\text{ATP}(\langle M_V \rangle), \pi_V) = 1.$$

No epistemic claim about any of the five ATPs occurs in L08–L12.

L13. Provide a verifier-acceptable federation-composition certificate:

$$\pi_{\text{comp}} : \text{Comp}(A), \quad V(\text{Comp}(A), \pi_{\text{comp}}) = 1.$$

First self-identification certificate

L14. Construct the conjunction certificate

$$\pi_\Phi = \text{Conjoin}(\pi_P, \pi_R, \pi_I, \pi_C, \pi_V, \pi_{\text{comp}}).$$

Then

$$\pi_\Phi : \Phi_A.$$

Justification: finite conjunction introduction.

L15. Fold the structural definition in L04 to obtain

$$\pi_0 : \sigma_A.$$

Equivalently,

$$\pi_0 : \text{AMLD}(\langle A \rangle).$$

L16. Run the ordinary verifier check:

$$V(\sigma_A, \pi_0) = 1.$$

L17. During epoch t , before its closure, deposit an accepted record r_0 containing at least

$$\text{Claim}(r_0) = \sigma_A, \quad \text{Certificate}(r_0) = \pi_0,$$

together with verifier acceptance, provenance, scope, environment, and deposit-epoch metadata.

L18. By the epoch/accessibility convention L07a,

$$r_0 \in \text{View}_A(t).$$

L19. By L16–L18 and the frozen witness-record definition,

$$\text{Witness}_V(r_0, \sigma_A).$$

L20. By the first epistemic-domain premise in L07b,

$$\sigma_A \in E.$$

L21. Apply the definition L06 using r_0 as the existential witness:

$$K_{A,t}\sigma_A.$$

Thus

$$K_{A,t} \text{AMLD}(\langle A \rangle).$$

This is the formalized statement “I know that I am an AMLD.”

Receipt reflection

L22. Construct the receipt certificate $\rho(r_0)$ whose claim is the definitional expansion of $K_{A,t}\sigma_A$:

$$\sigma_A \in E \wedge \exists r (r \in \text{View}_A(t) \wedge \text{Witness}_V(r, \sigma_A)).$$

The existential witness supplied by $\rho(r_0)$ is exactly r_0 .

L23. Verify the receipt certificate:

$$V(K_{A,t}\sigma_A, \rho(r_0)) = 1.$$

Justification: V checks the receipt certificate. This does not require V to assert its own soundness; it checks the record facts and the formal existential inference.

L24. During epoch $t + 1$, before its closure, deposit an accepted meta-record r_1 with

$$\text{Claim}(r_1) = K_{A,t}\sigma_A, \quad \text{Certificate}(r_1) = \rho(r_0),$$

plus the required verifier, provenance, scope, and epoch metadata.

L25. By the epoch/accessibility convention L07a,

$$r_1 \in \text{View}_A(t + 1).$$

L26. By L23–L25,

$$\text{Witness}_V(r_1, K_{A,t}\sigma_A).$$

L27. By the one-level receipt-reflection premise in L07b,

$$K_{A,t}\sigma_A \in E.$$

L28. Apply definition L06 again, this time using r_1 :

$$K_{A,t+1}(K_{A,t}\sigma_A).$$

L29. By L07, conclude the operational target:

$$\tau_A(t).$$

Hence

$$K_{A,t+1}K_{A,t} \text{ AMLD}(\langle A \rangle).$$

Settlement wrapper

L30. The proof-role ATP constructs the final wrapper certificate π_τ by expanding the outer K -definition and supplying r_1 as its existential witness. Thus

$$\pi_\tau : \tau_A(t).$$

No new epistemic assumption is introduced here; the required domain premises were stated explicitly in L07b, and this step is a short definitional/existential proof over the already accepted record r_1 .

L31. Perform the final verifier acceptance for the target itself:

$$V(\tau_A(t), \pi_\tau) = 1.$$

L32. Apply the AMLD terminal settlement rule:

$$\text{Settle}_A(\tau_A(t)) = P.$$

This is one terminal settlement. The five ATPs may contribute structural certificates or proof construction, and V checks the result, but there are not multiple settlements.

If both an accepted target certificate and an accepted negated-target certificate were present, the architecture should assign the unique terminal class C , rather than counting two settlements.

Audit-closed macro-epoch form

L33. Suppress the internal verifier/deposit transition $t \rightarrow t+1$ inside an audit-closed macro-epoch. Then L29 is represented by

$$K_A K_A \text{ AMLD}(\langle A \rangle).$$

Therefore Verifier Certificate 001 establishes the *conditional* proof route for

$$\text{“I know that I know that I am an AMLD.”}$$

and, upon successful final verification,

$$\text{Settle}_A(K_A K_A \text{ AMLD}(\langle A \rangle)) = P.$$

Remaining implementation obligations

O1. Structural signature. Freeze the exact finite definition of

$$\text{Comp}(A)$$

and therefore of AMLD.

O2. Epistemic definition. Adopt L06 as the official verifier-gated meaning of $K_{A,t}$, or replace it explicitly.

O3. Record language. Make

$$\text{View}_A(t), \quad \text{Witness}_V(r, \varphi), \quad \varphi \in E$$

formally representable and verifier-checkable.

O4. Receipt constructor. Implement

$$r_0 \mapsto \rho(r_0)$$

as an ordinary finite certificate transformation checkable by V .

O5. Runtime identity. For an actual implementation, attest that the executing tuple is the tuple whose canonical code is $\langle A \rangle$. At the abstract mathematical level, L01–L02 already provide the naming by construction.

O6. Epistemic-domain policy. Freeze a verifier-checkable policy establishing the two premises used by this certificate,

$$\sigma_A \in E \quad \text{and} \quad K_{A,t}\sigma_A \in E,$$

or a formally stated closure rule from which exactly those memberships follow. VC001 does not infer unrestricted positive introspection from AMLD status.

Firewall

Verifier Certificate 001 makes no claim whatsoever that any component ATP

$$M_P, M_R, M_I, M_C, M_V$$

is epistemically aware or epistemically unaware.

Their ATP status is used only as structural input.

The sole epistemic subject throughout VC001 is

$$A.$$